

ABB ACS580 GENERAL PURPOSE FREQUENCY CONVERTER
TECHNICAL DATASHEET & APPLICATION GUIDE - REF: ACS580-01-032A-4

1. PRODUCT OVERVIEW & DRIVING CAPABILITY

Series: ABB ACS580 Standard Industrial Drives

Model Code: ACS580-01-032A-4 (Wall-Mounted Single Drive)

Rated Power (Heavy Duty): 11.0 kW / 15.0 HP (Constant Torque)

Rated Power (Normal Duty): 15.0 kW / 20.0 HP (Variable Torque / Fans & Pumps)

Nominal Output Current: 32.0 A continuous

Maximum Overload Current: 48.0 A (150% overload for 60 seconds every 10 min)

Frame Size: R3 (Dimensions: 490 x 203 x 229 mm, Weight: 14.9 kg)

Enclosure Protection: IP21 Standard (UL Type 1), Optional IP55 (UL Type 12)

2. ELECTRICAL SUPPLY & OUTPUT CHARACTERISTICS

Mains Input Voltage Range: 3-Phase 380 V to 480 V AC (+10% / -15%)

Input Frequency: 50 / 60 Hz (+/- 5%)

Output Voltage: 0 to U_{input} (3-phase symmetrical)

Output Frequency Range: 0 to 500 Hz

Efficiency @ Nominal Rating: 98.2 %

Integrated EMC Filter: C2 Category (EN 61800-3 compliant for public low-voltage networks)

Braking Chopper: Built-in standard for R1, R2, R3 frames

3. CONTROL MODES & PROGRAMMING

Control Modes: Scalar Control, Sensorless Vector Control (Speed & Torque)

Speed Regulation Accuracy: +/- 0.5% nominal speed in vector mode

Analog Inputs: 2 x AI (0-10 V or 0/4-20 mA configurable)

Analog Outputs: 2 x AO (0/4-20 mA)

Digital Inputs: 6 x DI (24 VDC PNP/NPN, programmable)

Relay Outputs: 3 x Form C RO (250 VAC / 30 VDC, 2 A)

Safe Torque Off (STO): Dual-channel SIL3 / PL e certified according to EN ISO 13849-1

4. INDUSTRIAL COMMUNICATION & FIELDBUS

Embedded Fieldbus: Modbus RTU via RS-485 standard

Optional Adapter Modules: PROFINET (FPNO-21), EtherNet/IP (FEIP-21), EtherCAT (FECA-01), Modbus TCP (FMBT-21), CANopen (FCAN-01)